

```
!pip install -q groq

from groq import Groq

GROQ_API_KEY = "gsk_YbC5ywf1NCVLXo5WsNdYWGdyb3FYuUxdZM3h....."

client = Groq(api_key=GROQ_API_KEY)
```

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HITL stands for Human-In-The-Loop. In the context of LangGraph, a large language model, HITL refers to a process where human evaluation is integrated into the model's workflow.

Here's a simplified overview of how HITL works in LangGraph:

1. **Model generates response**: LangGraph generates a response to a given input or question.
2. **Human evaluation**: A human evaluator reviews the response to check its accuracy, relevance, and quality.
3. **Feedback loop**: The human evaluator provides feedback on the response, such as correcting errors, suggesting improvements, or highlighting areas of strength.
4. **Model updating**: The feedback is used to update and fine-tune the LangGraph model, so it can learn from its mistakes and improve its performance.
5. **Iteration**: Steps 1-4 are repeated, with the model generating new responses and human evaluators providing feedback, to continuously refine the model's output.

The HITL process in LangGraph helps to:

- * Improve the accuracy and relevance of the model's responses
- * Reduce errors and biases
- * Increase the model's ability to understand nuances and context
- * Enhance the overall quality of the model's output

By combining the strengths of both human evaluation and machine learning, HITL in LangGraph enables the development of more accurate and reliable AI models.

```
response = client.chat.completions.create(
    model="llama-3.3-70b-versatile",
    messages=[
        {
            "role": "user",
            "content": "Explain everything you know about Marwadi University."
        }
    ]
)
```

```
print(response.choices[0].message.content)
```

Marwadi University is a private university located in Rajkot, Gujarat, India. Here's an overview of what I know about the university:

History and Establishment: Marwadi University was established in 2016 by the Marwadi Education Foundation, a trust founded by the Marwadi community.

Academics: Marwadi University offers a wide range of undergraduate, postgraduate, and doctoral programs in various fields, including:

1. Engineering (B.Tech, M.Tech, Ph.D.)
2. Management (BBA, MBA, Ph.D.)
3. Law (BA LLB, B.Com LLB, LL.M, Ph.D.)
4. Science (B.Sc, M.Sc, Ph.D.)
5. Commerce (B.Com, M.Com, Ph.D.)
6. Arts (BA, MA, Ph.D.)
7. Pharmacy (B.Pharm, M.Pharm, Ph.D.)
8. Computer Applications (BCA, MCA, Ph.D.)

Infrastructure and Facilities: The university has a sprawling campus spread over 32 acres, with modern infrastructure and state-of-the-art facilities, including:

1. Well-equipped laboratories and workshops
2. Library with a vast collection of books, journals, and digital resources
3. Auditorium and conference halls
4. Sports facilities, including a cricket stadium, gymnasium, and swimming pool
5. Separate hostels for boys and girls
6. Canteen and food court
7. Medical facility and ambulance service

Research and Innovation: Marwadi University encourages research and innovation among its students and faculty members. The university has established several research centers and initiatives, including:

1. Marwadi University Incubation Center (MUIC)
2. Center for Innovation and Entrepreneurship (CIE)
3. Research and Development Cell (RDC)

Accreditation and Rankings: Marwadi University has received accreditation from various organizations, including:

1. National Assessment and Accreditation Council (NAAC) - 'B' Grade
2. National Accreditation Board for Education and Training (NABET)
3. The university has also been ranked among the top private universities in India by various ranking agencies, including The Times Higher Education and NIRF.

Student Life: Marwadi University has a vibrant student community, with various clubs, societies, and organizations that cater to the interests and talents of its students.

****Placement**:** Marwadi University has a dedicated placement cell that provides training and placement assistance to students. The
Overall, Marwadi University is a relatively new but rapidly growing institution that offers a wide range of academic programs, r